THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In the pattern of cube numbers $1, 8, 27, 64, \ldots$ the term after 343 is: 1

(a) 364 **(b)** 64 **(c)** 407 **(d)** 512

2. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 10th term is: 1

(a) 144 **(b)** 55 **(c)** 89 **(d)** 123

3. A complete graph has 6 points, each joined to every other point. How many lines are there? 1

(a) 30 **(b)** 21 **(c)** 36 **(d)** 15

4. The sum of the triangular numbers 28 and 36 is: 1

(a) 71 **(b)** 63 **(c)** 45 **(d)** 64

5. What is the sum of the first 30 natural numbers $1 + 2 + 3 + \cdots + 30$? 1

(a) 495 **(b)** 465 **(c)** 900 **(d)** 930

6. The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 21st one is: 1

(a) 462 **(b)** 420 **(c)** 42 **(d)** 441

7. In the pattern $1, 3, 6, 10, \ldots$ what is the 14th term? 1

(a) 120 **(b)** 105 **(c)** 91 **(d)** 196

8. In the pattern of cube numbers $1, 8, 27, 64, \ldots$ the term after 512 is: 1

(a) 729 **(b)** 593 **(c)** 81 **(d)** 536

9. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 11th term is: 1

(a) 89 **(b)** 144 **(c)** 233 **(d)** 199

10. A complete graph has 7 points, each joined to every other point. How many lines are there? 1

(a) 28 **(b)** 42 **(c)** 21 **(d)** 49

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum $1 + 3 + 5 + \ldots$ of the first 17 odd numbers without adding one by one. 2

12. Write the first 10 terms of the Virahanka sequence that starts with 1, 4. 2

13. How many lines are needed to join all pairs among 12 points (a complete graph)? 2

14. Find the sum $1 + 3 + 5 + \ldots$ of the first 20 odd numbers without adding one by one. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Add the triangular numbers 91 and 105. What do you notice about the answer? 3

16. Write the sum $1^3 + 2^3 + \cdots + 7^3$ and show it equals the square of $1 + 2 + \cdots + 7$. 3

17. Look at the sequence $7, 13, 19, 25, 31$. Find the pattern and the 10th term. 3

OR

Add the triangular numbers 105 and 120. What do you notice about the answer? 3

18. Write the sum $1^3 + 2^3 + \cdots + 8^3$ and show it equals the square of $1 + 2 + \cdots + 8$. 3

OR

Look at the sequence $13, 18, 23, 28, 33$. Find the pattern and the 10th term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A group of 10 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

OR

In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 9th and 10th terms are 55 and 89. Find the next three terms and add the first 9 terms. 5

20. A group of 11 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

OR

In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 10th and 11th terms are 89 and 144. Find the next three terms and add the first 10 terms. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND